Solving Polynomial Systems of using Linear Elimination

Ilay Menahem, Michael Shalyt, Ido Kaminer

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Abstract

1 Introduction

2 Preliminaries

Let K be a computable field and let $R = K[x_1, \ldots, x_n]$. The implementation considered in this paper uses $K = \mathbb{Q}$ and exact arithmetic. Geometric statements are made over an algebraic closure $\overline{K}$. For a finite set $F \subset R$, write

$$\langle F \rangle = \left\{ \sum_{f \in F} a_f f : a_f \in R \right\}, \qquad V(F) = \{a \in \overline{K}^n : f(a) = 0 \text{ for every } f \in F\}.$$

Thus, throughout the paper, a polynomial system is *satisfiable* if its zero set in $\overline{K}^n$ is nonempty. This convention is important: over an algebraically closed field, Hilbert's Nullstellensatz gives

$$V(F) = \varnothing \quad \Longleftrightarrow \quad 1 \in \langle F \rangle. \tag{1}$$

Consequently, consistency can be decided by an exact Gröbner-basis computation. The same equivalence does not hold for real feasibility without additional real-algebraic machinery; real solution sets are not considered here. We use standard terminology from computational commutative algebra as in [2, 3].

2.1 Gröbner bases and interreduction

Fix an admissible monomial order $\prec$ on R. For a nonzero polynomial f, denote its leading monomial by $\mathrm{LM}_{\prec}(f)$. A finite set $G \subset I$ is a Gröbner basis of an ideal I if

$$\langle \mathrm{LM}_{\prec}(g) : g \in G \rangle = \langle \mathrm{LM}_{\prec}(f) : f \in I \setminus \{0\} \rangle.$$

Polynomial reduction by G then supplies canonical normal forms when G is reduced. In particular, the reduced Gröbner basis of the unit ideal is $\{1\}$, so (1) is detected directly. Lexicographic orders are useful for elimination, whereas degree-reverse-lexicographic order is typically preferable for the consistency computations in the inner search. The correctness arguments below are independent of the chosen order.

Before factorization, the input generators may be *interreduced*: each generator is replaced by its remainder modulo the others, zero remainders are discarded, and this process is repeated to a fixed point. A single replacement $p_i \leftarrow p_i - qp_j$ preserves the generated ideal, since $p_i = (p_i - qp_j) + qp_j$. Hence sequential interreduction changes the presentation, but neither the ideal nor its zero set. This distinction matters because the factors exposed by preprocessing, and therefore the subsequent search space, may depend on the chosen generating set.

2.2 Factor hypergraphs and minimal covers

Let $P = \{p_1, \ldots, p_m\} \subset R$. After removing zero generators, write a complete factorization over K as

$$p_i = c_i \prod_{q \in C_i} q^{e_{i,q}}, \qquad c_i \in K^*, \quad e_{i,q} \geq 1, \tag{2}$$

where C_i is the set of distinct, nonconstant irreducible factors of p_i. Multiplicities play no role in a zero set and are therefore suppressed. A nonzero constant equation has $C_i = \varnothing$ and makes the system inconsistent, whereas a zero equation imposes no constraint and is omitted.

The family $\mathcal{C} = \{C_1, \ldots, C_m\}$ is a hypergraph on the universe $Q = \bigcup_i C_i$. A set $T \subseteq Q$ is a *factor cover*, or equivalently a transversal (hitting set) of $\mathcal{C}$, if $T \cap C_i \neq \varnothing$ for every i. It is *minimal* if none of its proper subsets is a factor cover. We denote the family of minimal factor covers by $\mathrm{Tr}(\mathcal{C})$. These are inclusion-minimal, not necessarily minimum-cardinality, transversals [1].

The following elementary identity is the basis of the outer branching method.

Lemma 1 (Factor-cover decomposition) *For the factorization in* (2)*,*

$$V(P) = \bigcup_{T \in \mathrm{Tr}(\mathcal{C})} V(T). \tag{3}$$

Equivalently,

$$\sqrt{\langle P\rangle} = \bigcap_{T\in \mathrm{Tr}(\mathcal{C})} \sqrt{\langle T\rangle}.$$

Proof 1 *If $a \in V(T)$ and T hits every C_i, then some factor of every p_i vanishes at a; hence $a \in V(P)$. Conversely, if $a \in V(P)$, the set $Z_a = \{q \in Q : q(a) = 0\}$ hits every C_i. Since the hypergraph is finite, Z_a contains an inclusion-minimal transversal T, and $a \in V(T)$. The ideal identity follows from the correspondence between finite unions of algebraic sets and intersections of their radical ideals.*

A factor cover need not define an irreducible component: $\langle T\rangle$ may itself be reducible, and different covers may define nested or overlapping subvarieties. Accordingly, the method computes structured subvarieties whose union is $V(P)$, rather than claiming an irredundant primary decomposition. Restricting the enumeration to minimal covers loses no points, because if $T' \supset T$, then $V(T') \subseteq V(T)$.

2.3 Learned inconsistency and structure hints

An *unsatisfiable core* of a finite constraint set $F \subset R$ is a subset $U \subseteq F$ for which $V(U) = \varnothing$, equivalently $1 \in \langle U\rangle$. It is minimal if every proper subset is satisfiable. Unsatisfiability is monotone under adding constraints:

$$U \subseteq F' \quad \text{and} \quad V(U) = \varnothing \quad \Longrightarrow \quad V(F') = \varnothing. \tag{4}$$

Thus a learned core rules out every later candidate containing it. Dually, if $F' \subseteq F$ and $V(F) \neq \varnothing$, then $V(F') \neq \varnothing$. Core extraction may use a Gröbner-basis certificate and subsequently delete redundant constraints; the method section specifies the refinement procedure.

A *structure hint* is an optional polynomial equation $h = 0$. Given a finite hint set $H \subset R$, a branch selects some $A \subseteq H$ and studies $P \cup A$. Examples include sparsity constraints $x_i = 0$ and affine relations $a_1x_1 + \cdots + a_nx_n = b$. Since

$$A \subseteq A' \quad \Longrightarrow \quad V(P \cup A') \subseteq V(P \cup A),$$

an unsatisfiable hint core $U_H \subseteq H$, characterized by $V(P \cup U_H) = \varnothing$, prunes every hint branch containing U_H. Conversely, a satisfiable hinted branch defines a subvariety of $V(P)$ with the requested structure.

Finally, for a satisfiable candidate F, its geometric dimension means the Krull dimension of $R/\sqrt{\langle F\rangle}$, equivalently the dimension of $V(F)$. It can be read from the leading-monomial ideal of a Gröbner basis; radical computation is not required because an ideal and its radical have the same dimension [2].

3 Method

4 Variable Zeroing

5 Experiments

6 Conclusion

References

[1] Claude Berge. *Hypergraphs: Combinatorics of Finite Sets*. North-Holland, Amsterdam, 1989.

[2] David A. Cox, John Little, and Donal O'Shea. *Ideals, Varieties, and Algorithms: An Introduction to Computational Algebraic Geometry and Commutative Algebra*. Springer, Cham, 4 edition, 2015.

[3] Martin Kreuzer and Lorenzo Robbiano. *Computational Commutative Algebra 1*. Springer, Berlin, 2000.

A Appendix